Robotics

(Course viewgraphs)

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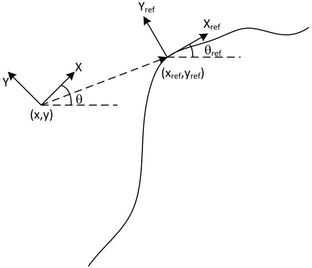
A Linear Model For the Unicycle Trajectory Tracking I

- Assume that the reference trajectory to track is generated by a virtual unicycle robot



$$\dot{x}_{ref} = \cos(\theta_{ref})v_{ref}$$
$$\dot{y}_{ref} = \sin(\theta_{ref})v_{ref}$$
$$\dot{\theta}_{ref} = \omega_{ref}$$

- The tracking errors in the world frame are

$$^{W}e_x = x_{ref} - x$$
$$^{W}e_y = y_{ref} - y$$
$$^{W}e_\theta = \theta_{ref} - \theta$$

A Linear Model For the Unicycle Trajectory Tracking II

- Assuming that the reference trajectory was generated by a unicycle, under some reference controls v_{ref} and ω_{ref},

$$
\begin{aligned}
{}^{W}\dot{e}_x &= \dot{x}_{ref} - \dot{x} = \cos(\theta_{ref})v_{ref} - \cos(\theta)v \\
{}^{W}\dot{e}_y &= \dot{y}_{ref} - \dot{y} = \sin(\theta_{ref})v_{ref} - \sin(\theta)v \\
{}^{W}\dot{e}_\theta &= \dot{\theta}_{ref} - \dot{\theta} = \omega_{ref} - \omega
\end{aligned}
$$

- The same tracking errors in the robot frame are

$$
\begin{bmatrix} {}^{B}e_x \\ {}^{B}e_y \\ {}^{B}e_\theta \end{bmatrix} = \begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 \\ -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} {}^{W}e_x \\ {}^{W}e_y \\ {}^{W}e_\theta \end{bmatrix}
$$

A Linear Model For the Unicycle Trajectory Tracking III

- The dynamics of the error is given by

$$
{}^{B}\dot{e} = \begin{bmatrix} -\sin(\theta) & \cos(\theta) & 0 \\ -\cos(\theta) & -\sin(\theta) & 0 \\ 0 & 0 & 0 \end{bmatrix} \dot{\theta}\, {}^{W}e + \begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 \\ -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} {}^{W}\dot{e}
$$

- Since

$$
{}^{W}e = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} {}^{B}e
$$

which can then be replaced in the above equation

A Linear Model For the Unicycle Trajectory Tracking IV

- Leading to

$$
{}^{B}\dot{e} = \begin{bmatrix} 0 & \omega & 0 \\ -\omega & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} {}^{B}e + \begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 \\ -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} {}^{W}\dot{e}
$$

A Linear Model For the Unicycle Trajectory Tracking V

- Replacing ${}^{W}\dot{e}$ by the adequate error expression

$$
\begin{aligned}
{}^{B}\dot{e} &= \begin{bmatrix} 0 & \omega & 0 \\ -\omega & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} {}^{B}e + \begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 \\ -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(\theta_{ref})v_{ref} - \cos(\theta)v \\ \sin(\theta_{ref})v_{ref} - \sin(\theta)v \\ \omega_{ref} - \omega \end{bmatrix} \\
&= \begin{bmatrix} 0 & \omega & 0 \\ -\omega & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} {}^{B}e + \begin{bmatrix} \cos(e_\theta)v_{ref} - v \\ \sin(e_\theta)v_{ref} \\ \omega_{ref} - \omega \end{bmatrix} \\
&= \begin{bmatrix} 0 & \omega & 0 \\ -\omega & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} {}^{B}e + \begin{bmatrix} 0 \\ \sin(e_\theta) \\ 0 \end{bmatrix} v_{ref} + \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(e_\theta)v_{ref} - v \\ \omega_{ref} - \omega \end{bmatrix}
\end{aligned}
$$

- Also, linearizing $\sin(e_\theta)$ around the equilibrium $e_\theta = 0$ means that ω can be replaced by ω_{ref}

A Linear Model For the Unicycle Trajectory Tracking VI

- Also,

$$
{}^B\dot{e} = \begin{bmatrix} 0 & \omega_{ref} & 0 \\ -\omega_{ref} & 0 & v_{ref} \\ 0 & 0 & 0 \end{bmatrix} {}^B e + \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_{ref} - v \\ \omega_{ref} - \omega \end{bmatrix}
$$

- A further simplifying assumption is often that $\omega_{ref} = cte$ (making the system time invariant)

Linear Control I

- The controllability matrix of a linear system $\dot{x} = Ax + Bu$, with $x \in \mathbb{R}^3$ is

$$\Gamma_c = \left[B \,|\, AB \,|\, A^2 B \right]$$

- Standard results from control theory for linear systems ensure that if $\text{rank}(\Gamma_c) = 3$ then the system is controllable

- For the current model

$$\text{rank}(\Gamma_c) = \begin{cases} 2 & \text{if } v_{ref} = \omega_{ref} = 0 \\ 3 & \text{otherwise} \end{cases}$$

- Therefore, if the reference vehicle is not stopped just use a standard state space law,

$$\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = - \begin{bmatrix} k_{11} & k_{12} & k_{13} \\ k_{21} & k_{22} & k_{23} \end{bmatrix} \begin{bmatrix} e_x \\ e_y \\ e_\theta \end{bmatrix}$$

Linear Control II

- The performance can be estimated from the position of the poles of the closed loop system

- Let the overall system be represented in a compact form as

$$\dot{e} = Ae + Bu$$

$$u = -Ke$$

  The poles of the closed-loop system are the eigenvalues of the matrix $A - BK$, i.e., the roots of $\det(sI - (A - BK)) = 0$

- Let the poles of the closed-loop system be represented as the solutions of the 3rd order polynomial

$$(s + a)(s^2 + 2\xi\omega_n s + \omega_n^2) = 0$$

  where the complex poles are responsible for oscillating behavior

Linear Control III

- A possible control law is

$$u = \begin{bmatrix} -K_1\, e_x \\ -K_2\, \mathrm{sgn}(v_{ref})\, e_y - K_3\, e_\theta \end{bmatrix}$$

which leads to

$$K_1 = 2\xi\omega_n \quad K_2 = \frac{\omega_n^2 - \omega_{ref}^2}{|v_{ref}|} \quad K_3 = 2\xi\omega_n$$

Non-Linear Control I

- An alternative control law is

$$u = \begin{bmatrix} -K_1(v_{ref}, \omega_{ref})\, e_x \\ \\ -K_2 v_{ref} \frac{\sin(e_\theta)}{e_\theta} e_y - K_3(v_{ref}, \omega_{ref}) e_\theta \end{bmatrix}$$

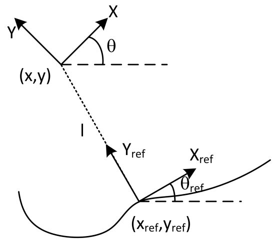
  where

  - $K_1(v_{ref}, \omega_{ref})$, $K_3(v_{ref}, \omega_{ref})$ are strictly positive continuous functions in $\mathbb{R} \times \mathbb{R}^2 \backslash \{(0,0)\}$
  - $K_2 > 0$ is a constant

- It can be proven that the above control law globally asymptotically stabilizes $e = 0$

Hint: Use, for example, the Lyapunov function $V = \frac{1}{2} e^T e$

Non-Linear Control II



Re-parameterizing the state space to

$$(s, l, \tilde{\theta})$$

where

- s is the curvilinear distance along the path
- l is the distance between the origins of the real and virtual robots frames
- $\tilde{\theta} = \theta_{ref} - \theta$ is the orientation error

- The error dynamic for the unicycle comes

$$\dot{s} = \frac{v\ \cos(\tilde{\theta})}{1 \mp c(s)\ l}$$

$$\dot{l} = v\ \sin(\tilde{\theta})$$

$$\dot{\tilde{\theta}} = \omega - \frac{v\ \cos(\tilde{\theta})\ c(s)}{1 \mp c(s)\ l}$$

where $c(s)$ is the curvature of the reference path, measured at the origin of the frame attached to the virtual robot

Non-Linear Control III

- Linearizing the two last equations around the equilibrium $(l, \tilde{\theta}) = (0, 0)$

$$\dot{l} = v\,\tilde{\theta}$$

$$\dot{\tilde{\theta}} = u$$

with $u = \omega - \frac{v\cos(\tilde{\theta})c(s)}{1 \mp c(s)\,l}$

- A linear stabilizing law is

$$u = -K_2 v\, l - K_3 |v| \tilde{\theta}$$

with $K_2, K_3 > 0$

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Non-Linear Control IV

- A non-linear law that also asymptotically stabilizes $(l, \tilde{\theta}) = (0, 0)$,

$$u = -K_2 v\, l\, \frac{\sin(\tilde{\theta})}{\tilde{\theta}} - K(v)\tilde{\theta}$$

with $K_2 > 0$ and $K(v)$ a continuous, strictly positive, function, provided that

$$l(0)^2 + \frac{1}{K_2}\tilde{\theta}(0)^2 < \frac{1}{\limsup c(s)^2}$$

Car-robot control I

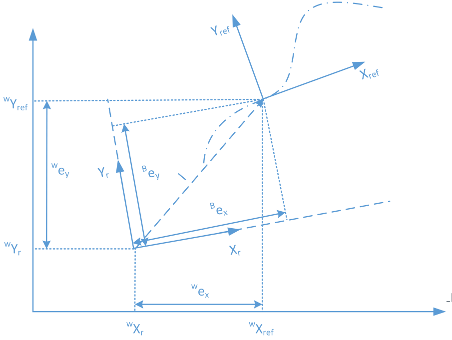
$$\begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{\phi} \end{bmatrix} = \begin{bmatrix} \cos(\theta) & 0 \\ \sin(\theta) & 0 \\ \tan(\phi)/L & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v \\ \omega_s \end{bmatrix}$$

$${}^{w}e = [x_{ref} - x, y_{ref} - y, \theta_{ref} - \theta]$$

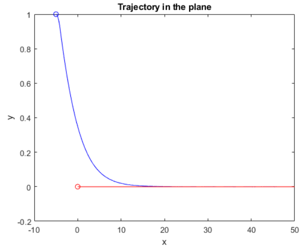
$${}^{b}e = \begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 \\ -\sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} {}^{w}e$$

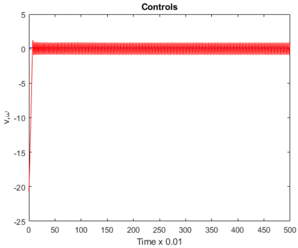
$$v = K_v \, {}^{b}e_x$$

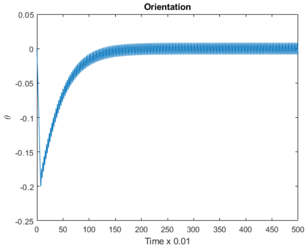
$$\omega_s = K_s \, {}^{b}e_\theta + K_l \, {}^{b}e_y$$



Car-robot control II







$K_v = 0.03$

$K_s = 100$

$K_l = 1$

$h = 0.01$

$|\phi| < \pi/8$

$L = 2.2$

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Controlling a bicycle I

A simplified kinematics model is given by

$$\dot{x} = \cos(\psi) v$$

$$\dot{y} = \sin(\psi) v$$

$$\dot{\psi} = \frac{\tan(\delta \; \cos(\gamma)) \; v}{w \; \cos(\phi)}$$

where ψ is the orientation in the world frame, δ is the steering angle, γ the rake angle (a constant),ϕ is the leaning angle (which can be assumed a constant for the purpose of a kinematic modeling) and w the wheelbase distance (also a constant under simplifying assumptions)

The control law used is

$$\delta = -K_1 \, v \, l + K_2 |v| \, (\psi_{ref} - \psi)$$

where

- l is the signed distance defined by the projection of the position of the bicycle onto the reference path,

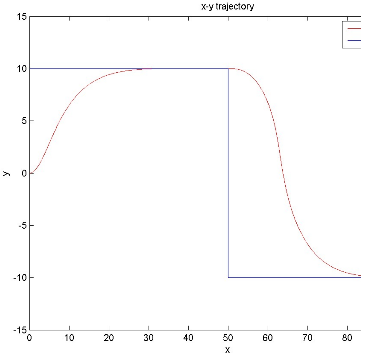
Controlling a bicycle II

- v is the linear velocity, ψ, ψ_{ref} are, respectively, the orientation of the bike and the corresponding reference

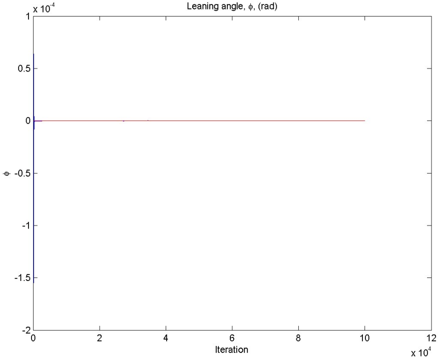
- K_1, K_2 are tuning constants

- It can be proven that this control law drives the distance error l and the orientation error $\psi_{ref} - \psi$ both asymptotically to zero (Sequeira, de Vittori,2012, "Motorbike Modeling and Control")

Controlling a bicycle III



Trajectory



Leaning angle

Nonlinear control law with $K_1 = 1$, $K_2 = 10$

Model Predictive Control I

- The idea is to predict the effect of the available controls in a finite horizon
- At each instant k, for each point in the horizon compute the cost

$$J(k) = \sum_{i=0}^{N_y} w_i \left(y_{ref}(k+i) - y(k+i)\right)^2 + w_u \sum_{i=0}^{N_u} \|u(k+i)\|^2$$

where

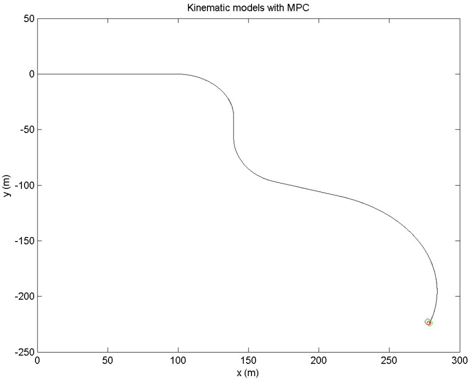
- u is the control vector
- y is the output
- y_{ref} is the output reference
- w_i are the output error weights
- w_u is the weight on the control effort
- N_u is the control horizon

Model Predictive Control II

  - N_y is the output error horizon

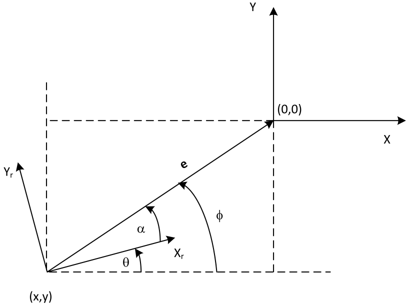
- Compute the sequence of controls $u(k), u(k+1), \ldots, u(k+N_u)$ that minimizes $J(k)$
- Apply $u(k)$
- Repeat the whole process for $k+1$
- If $y(u)$ is differentiable optimization tools can be used, e.g., Karush-Kuhn-Tucker conditions
- Otherwise/alternatively, tree-search techniques work in the general situation

Model Predictive Control III



- Same bicycle as before
- Discrete set of controls, $\delta = \{-1, -0.5, -0.2, 0, 0.2, 0.5, 1\}$

Position tracking I



- $e = \|\mathbf{e}\| = \sqrt{x^2 + y^2}$ is the length of the error vector $\mathbf{e} = (0,0) - (x,y)$
- $\phi = \text{atan2}(-y, -x)$
- $\alpha = \phi - \theta$

- The temporal evolution of the variables that represents the problem, e, ϕ, α is given by

$$\dot{e} = \frac{x}{e}\dot{x} + \frac{y}{e}\dot{y} = -\cos(\phi)\,\dot{x} - \sin(\phi)\,\dot{y}$$

$$\dot{\phi} = \frac{1}{e}\left(-\frac{y}{e}\dot{x} + \frac{x}{e}\dot{y}\right) = \frac{1}{e}\left(-\sin(\phi)\,\dot{x} - \cos(\phi)\,\dot{y}\right)$$

$$\dot{\alpha} = \dot{\phi} - \dot{\theta}$$

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Position tracking II

- To include the unicycle robot just replace $\dot{x}, \dot{y}, \dot{\theta}$ by the corresponding expressions

$$\dot{e} = -\cos(\alpha)\, v$$

$$\dot{\phi} = \frac{\sin(\alpha)}{e}\, v$$

$$\dot{\alpha} = -\omega + \frac{\sin(\alpha)}{e}\, v$$

- A possible control law is,

$$v = v_{\max} \tanh(K_1 e)$$

$$\omega = v_{\max} \left(\left(1 + K_2 \frac{\phi}{\alpha}\right) \frac{\tanh(K_1 e)}{e} \sin(\alpha) + K_3 \tanh(\alpha) \right), \quad K_1, K_3 > 0$$

which can be shown to asymptotically stabilize $e = 0$

Hint: Use the Lyapunov function $V \equiv \frac{1}{2} e^2 + \frac{1}{2}(K_2 \phi^2 + \alpha^2)$, with $K_2 > 0$

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Additional bibliography for robot control

- "Tracking Control of Unicycle-Modeled Mobile Robots Using a Saturation Feedback Controller". Ti-Chung, Kai-Tai Song, Ching-Hung Lee, Ching-Cheng Teng. IEEE Transactions on Control Systems Technology 9(2), March 2001

- "Modeling and Adaptive Control of an Omni-Mecanum-Wheeled Robot". Lih-Chang Lin, Hao-Yin Shih. Intelligent Control and Automation 4:166-179, 2013

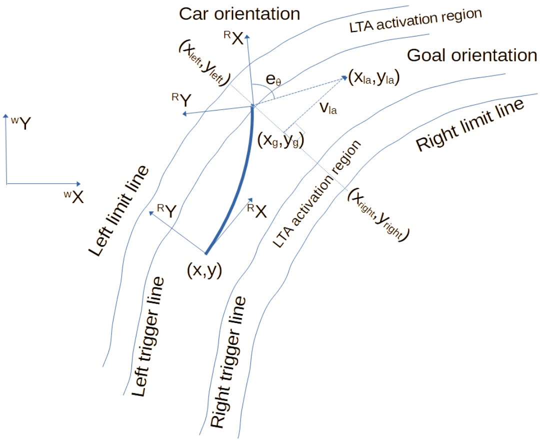
- "Bounded Smooth Time Invariant Motion Control of Unicycle Kinematic Models". Youngshik Kim, Mark A. Minor. Procs. of the 2005 IEEE Int. Conf. on Robotics and Automation, April 2005

- "Two Observer-Based Tracking Algorithms for a Unicycle Mobile Robot" J. Kakubiak, E. Lefeber, K. Tchon, H. Nijmeijer. Int. J. Appl. Math. Comp. Sci, 12(4):513-522, 2002

- "The Path Following Control of a Unicycle Based on the Chained Form of a Kinematic Model Derived with Respect to the Serret-Frenet Frame". J. Plakonska. Procs. of the 17th Int. Conf. on Methods and Models in Automation and Robotics (MMAR 2012), 27-30 Aug, 2012

LTA control I

- Not an autonomous car; it's not a trajectory following problem
- The ideas in the previous slides can be adjusted to the LTA problem



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LTA control II

- The goal orientation can be defined after a look-ahead point; a reasonable control can be

$$w = K\, e_\theta$$

- Finding the projections $(x_{left}, y_{left}), (x_{right}, y_{right})$ depends on the sensors available

$$(x_{left}, y_{left}) = \min \|\text{left line} - (x, y)\|$$

$$(x_{right}, y_{right}) = \min \|\text{right line} - (x, y)\|$$

- The look ahead goal point can be defined as

$$(x_g, y_g) = \max \left\{ \|(x_{left}, y_{left}) - (x_g, y_g)\| + \|(x_{right}, y_{right}) - (x_g, y_g)\| \right\}$$

$$(x_{la}, y_{la}) = (x_g, y_g) + v_{la}$$

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